Automated Gold & Forex Trading System

Decision Brief for Non-Technical Stakeholders

What This Is

This project builds a **24/7 automated trading system for Gold and Forex** that replicates how a professional trading team operates — five specialists working together on every single trade decision before any money is committed to a trade.

Instead of one "black-box" algorithm making isolated decisions, the system enforces debate and checks at every step:

The Five-Person Team:

1. **The Brain (AI Council)** — Three analytical voices score each trading opportunity: a bull (looks for reasons to buy), a bear (looks for reasons to sell), and a risk-management voice (watches for danger signs like too-wide stop-losses, approaching economic news, or market turmoil). A trade only proceeds if it passes a majority vote and passes the risk-manager's veto gate. When a decision is borderline or high-stakes, the system escalates to Claude AI for a human-level sanity check before acting.
2. **The Shield (Portfolio Checkpoint)** — Before the Council's idea becomes an order, this checkpoint asks: is this trade good enough on its own (win/loss ratio)? Would it duplicate risk we already have? Does it fit the current portfolio direction? If not, it blocks the trade with a logged reason.
3. **The CFO (Money Manager)** — Given an approved trade idea, this role calculates the right position size: How much of the account should we risk? Adjust for current market volatility. Size the position so even a losing trade doesn't blow up our capital. It also enforces a daily loss limit — if we've had a bad day, no more new trades until tomorrow.
4. **The Watchman (Position Monitor)** — Once a trade is open, this role actively manages it: trails the stop-loss to lock in gains, exits early if the market structure that justified the trade breaks, protects profits once they form. It works alongside a broker-side stop-loss (a hard safety net independent of our system).
5. **The Auditor (Performance Reviewer)** — Each day, reviews what happened: which trades worked, which didn't, why, what we learned. Most importantly: decides which strategies

have proven themselves enough to risk real money. Nothing moves from testing to live trading without the Auditor's sign-off.

> **Core Safety Philosophy:**
> Nothing goes live with real capital until it has *proven itself three times over*:
>
> - Passes historical simulation (backtesting) on data it wasn't trained on
> - Runs weeks in real-time paper trading (live market conditions, zero real money at risk)
> - Starts with tiny real trades and ramps up gradually — never a sudden jump to full position size

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How It Gets Built

This is not built all at once. Instead, it's built in **five plain-language stages**, each proven to work before the next begins:

Stage 1: Foundation & Data (Phases 0–1)

Set up the basic plumbing: get Python and the trading platform talking, log into real-time market data, download historical price data for backtesting.

> **Proof point:** The system can see and record live market data.

Stage 2: Proof of Concept — Tiny Test Trades (Phases 2–3)

Implement one very simple trading rule and run it in shadow-mode on the broker's demo account (no real money, no risk). The full pipeline — idea generation, risk checks, position sizing, order placement — runs end-to-end, but trades are throttled and logged for manual verification.

> **Proof point:** The system can place trades, broker infrastructure works, nothing crashes.

Stage 3: Add Intelligence & Optimization (Phases 4–8)

Build the backtest engine so we can test strategies on years of historical data. Implement the full five-person team: Council (multi-voice debate), Shield (portfolio-level checks), CFO (intelligent sizing), Watchman (trailing stops). Each piece is tested independently and together.

> **Proof point:** Strategies demonstrably profit in historical simulation without overfitting.

Stage 4: Extended Real-Time Testing with No Real Money (Phase 9)

Run the full system in paper-trading mode for weeks. Live market data, real conditions, but every trade is simulated — no capital at risk. The Auditor produces daily reports. We validate that the strategy holds up in reality.

> **Proof point:** The system performs as expected when facing real market conditions, surprises are understood.

Stage 5: Careful Rollout to Real Trading (Phases 10–11)

Deploy to a VPS (a small cloud server running 24/7). First trades are tiny (0.25% of account risk per trade). Gradually increase to full size only after live performance confirms the strategy works with real money.

> **Proof point:** Real trading works, monitoring/alerts work, strategy survives first contact with the live market.

Why this approach works: Each stage catches errors before they become expensive. If the backtest looks good but paper trading reveals a flaw, we fix it without touching real capital. Most problems surface in stages 2–4, not after months of build time.

Cost Breakdown

All costs are for **infrastructure** — the tools and cloud services the system runs on. This is *separate* from trading capital (the money being traded with) and trading losses (the risk of trading itself).

Item	Cost	Notes
Trading platform (MT5)	Free	Provided by broker
Python & development	Free	Open-source
Cloud server (VPS)	$10–30/month	Entry to mid-tier; premium servers ($60–190/month) aren't needed for our decision frequency
AI (Claude API)	$5–30/month	Used for trade debate escalation and daily performance summaries; starts Phase 10
Market data & news	Free	Free sources initially; premium feeds optional later
Monitoring alerts	Free	Standard integrations (email, Telegram, Discord)
Total monthly infrastructure	**~$15–60/month**	**Once fully live (Phase 5)**

Important distinction:

- **Infrastructure cost** (~$15–60/month) is the operational overhead, separate from trading costs.
- **Trading capital** is the money being traded with — that's a separate decision and risk.
- **Trading losses** are the money lost from bad trades — that's the risk of the strategy, managed by daily loss limits, forced ramp, and the Auditor's gating.

AI Provider Options (Instead of Claude)

The "AI (Claude API)" line above is only used for one narrow job: double-checking borderline trade decisions and writing the daily performance summary. It is **not** required for the system to work — the five-person team's rule-based decision-making runs on its own without any AI API. Because this piece is isolated to one small part of the system, switching providers later is easy — it doesn't affect anything else that's been built.

Option	Est. Cost/Month	Trade-off
No AI at all — team runs on rules only	**$0**	Recommended starting point. The team's math-and-rules-based decisions are fully functional without any AI double-check. Add AI later only if there's a proven need.
Budget AI options (Groq, or similar low-cost providers)	~$1–5	Cheapest paid option. Lower judgment quality — worth testing before trusting it on real trades.
Google's AI (Gemini)	~$2–10	Often has a free usage tier that may cover this system's light, occasional use entirely.
OpenAI (makers of ChatGPT)	~$2–15	Comparable quality range to Claude, similar cost.
Claude (Anthropic) — current plan	~$5–30	The default recommendation — favors a stronger AI for "should we risk real money on this" decisions over marginal savings.
Run an AI model ourselves (no per-use fee)	$0 extra fee, but a bigger/pricier server	Avoids AI subscription costs but needs a more powerful (and more expensive) cloud server, plus more setup work. Not recommended for a first build.

> **Bottom line:** this is not a decision that needs to be made now. The recommended path is to build and prove the system with **no AI subscription at all** (Stages 1–4), and only decide on an AI provider once there's a real track record showing it's needed — at that point, pick based on actual usage, not an estimate.

What's Still Undecided (Non-Blockers)

A handful of choices will be made as each build stage approaches. These are **not** holding up the project:

- **Which broker & account** — details on trading fees, contract sizes, specific platform rules. Chosen before Phase 3.
- **Which VPS provider** — cloud hosting choice (general-purpose vs. forex-specialized). Chosen before Phase 11.
- **News data source** — free economic calendar or paid real-time news feed. Decided as Phase 2 starts.
- **AI budget ceiling** — determines whether we use cost-optimized Claude or premium tier for edge cases. Set before Phase 10.

- **Technical thresholds** — specific win-rate requirements for "proven" strategy, daily loss limit amount, circuit-breaker behavior. These are written into configuration and can be adjusted.

None of these are architectural problems or surprises. They're choices that depend on broker selection or budget preferences.

Risk Acknowledgment

Automated trading carries real risks:

- Strategies can fail on unseen market conditions despite backtesting.
- Execution errors, broker platform issues, or connectivity gaps can cause losses.
- Market events (flash crashes, gaps) can exceed even carefully-sized stop-losses.

How this system mitigates those risks:

- The five-person-team approach catches logical errors that a solo algorithm would miss.
- No decision is trusted without proof: backtest, paper trading, then live ramp.
- Hard broker-side stop-losses (independent of our monitoring) protect every position.
- Daily loss limits prevent "death spiral" days from compounding.
- Full audit trails make every decision reviewable — when something goes wrong, we know why.
- The Auditor controls which strategies touch real money, using objective historical criteria.

This is not "fire and forget." It's continuous automated trading with multiple layers of human-like reasoning and mechanical safety checks.

Next Steps

To greenlight:

- Confirm budget for infrastructure (~$15–60/month).
- Confirm willingness to invest development time through Stage 2 (proof the system can place trades without crashing).

- Identify broker preference (needed by Phase 3 to finalize platform integration details).

To proceed, this document should be reviewed alongside:

- The technical specification (`spec.md`) for architecture details if needed.
- The verification summary (final section of spec) listing concrete proof-points before any live capital is committed.
